

Elastic MCP 及 Agentic AI - 构建可信、 上下文感知能力的搜索

演讲嘉宾：刘晓国

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01

搜索市场演变

搜索市场变迁



基于搜索的应用

支持广泛的使用场景

- 文档与业务数据搜索
- 终端用户 App 体验（例如 抖音，滴滴，美团等）
- 内部应用（大型组织中有数百个app）
- 数据库分流
- 电子商务
- 可扩展、灵活的查询与相关性

传统使用场景



向量与混合搜索

提升相关性，并为 GenAI / RAG 奠定基础

- 语义搜索提升现有的 app 和使用场景
- GenAI / RAG 工作流需要 Vector DB+
- 现有数据更有价值，并推动新的使用场景
- 终端用户对开发者的期望不断提高
- 更好的相关性带来更好的结果

快速增长



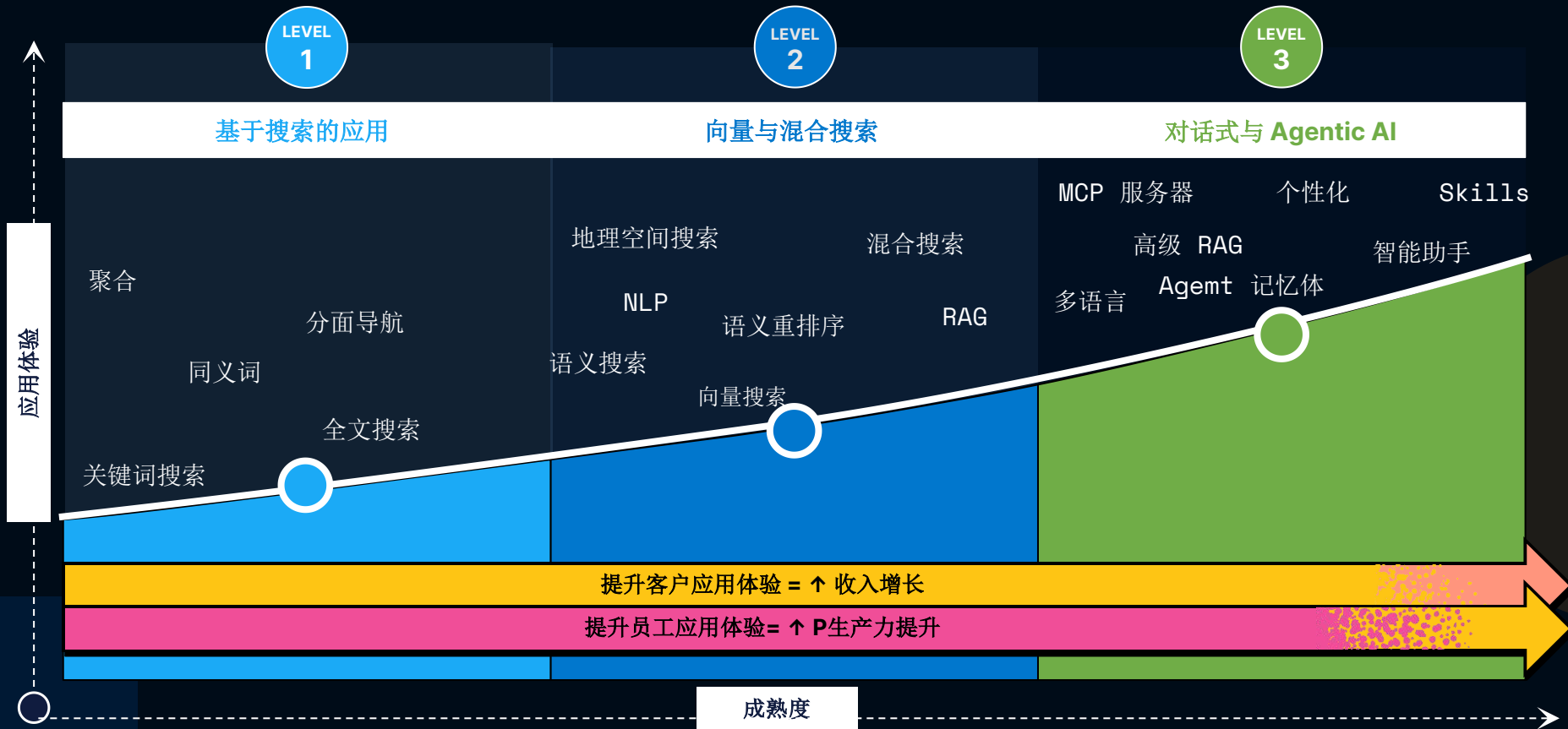
对话式与 Agentic AI

基于搜索构建的体验

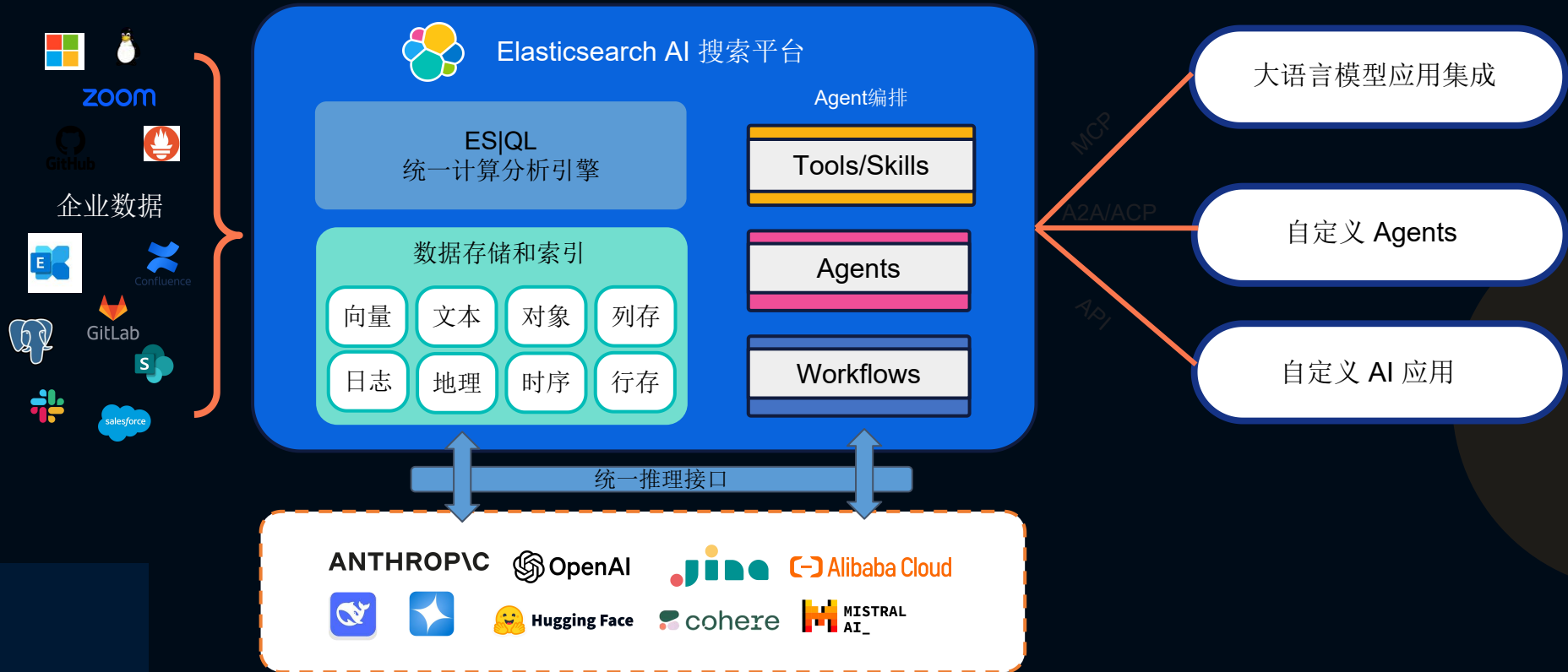
- 聊天 / 对话式体验
- 基于用户参与和自动化的 AI 驱动工作流
- 在现有数据源之上启用新的应用类别
- 期望直接由终端用户驱动
- 需要卓越的相关性，尤其是在长尾查询中

早期采用

客户旅程



Elasticsearch AI 搜索平台整体架构



Elasticsearch 最强的混合搜索引擎

一个 API 支持最复杂的混合搜索

数据写入

- AI 生态系统
- 200+ 数据连接器
- RAG Playground
- API 写入数据
- Eland 模型导入
- 第三方模型整合

数据类型

- Dense vectors
- Sparse vectors
- Numerics
- Spatial
- Temporal
- Boolean
- Terms
- Objects
- Aliases

召回

- Semantic Text
- BM25f
- kNN
- Rank features
- Scripts
- Rules
- Filters
- Suggesters
- Aggregations
- Pre filtering

重排

- Jina Rerank
- Rank fusion (RRF)
- Linear retrieval
- Reranking
- Rescoring
- Post filtering

其它

- Highlighting
- Sorting
- Pagination

优化

- Learning to rank
- Engagement data
- Ranking evaluation
- Benchmarking
- Profiling

AI 整合

- 云 AI 基础设施
- 自部署 AI
- NLP 模型直接部署到 ES
- 第三方开源 AI 工具
- 第三方商业 AI 方案

AI 搜索平台

- ✓ 可组合的模板、插件、脚本
- ✓ 弹性自动扩展、高可用性、快照
- ✓ 部署、索引、文档、字段的权限管理，基于角色访问控制 和 基于属性访问控制
- ✓ 无缝搜索跨索引、部署、区域

提问的方式变了...

搜索也应该跟着改变

在单个 ES|QL 查询中，自然语言查询（例如“**What is ES|QL?**”）可以分阶段组合：

1. 初始结果检索
2. RRF 合并 + 语义重排序结果
3. 对前 n 个结果进行 LLM 摘要

在 Elasticsearch 中，RAG 就像执行一个 ES|QL 查询一样简单

“What's
ES|QL?”

```
FROM search-index METADATA _score
| FORK (WHERE semantic_content:"what's esql?"
        | SORT _score DESC)
        (WHERE content:"what's esql?"
        | SORT _score DESC)
| FUSE
| RERANK "what's esql" ON content WITH "elastic_rerank"
| LIMIT 6
| COMPLETION CONCAT("Summarize :", content) WITH
"elasticllm" AS summary
```


将企业数据引入 agents 是一个挑战

Chat Experiences

 Claude  ...

Custom agents

  LangChain ...



相关性 & 可靠性



可扩展性 & 效率



集成 & 可延续性

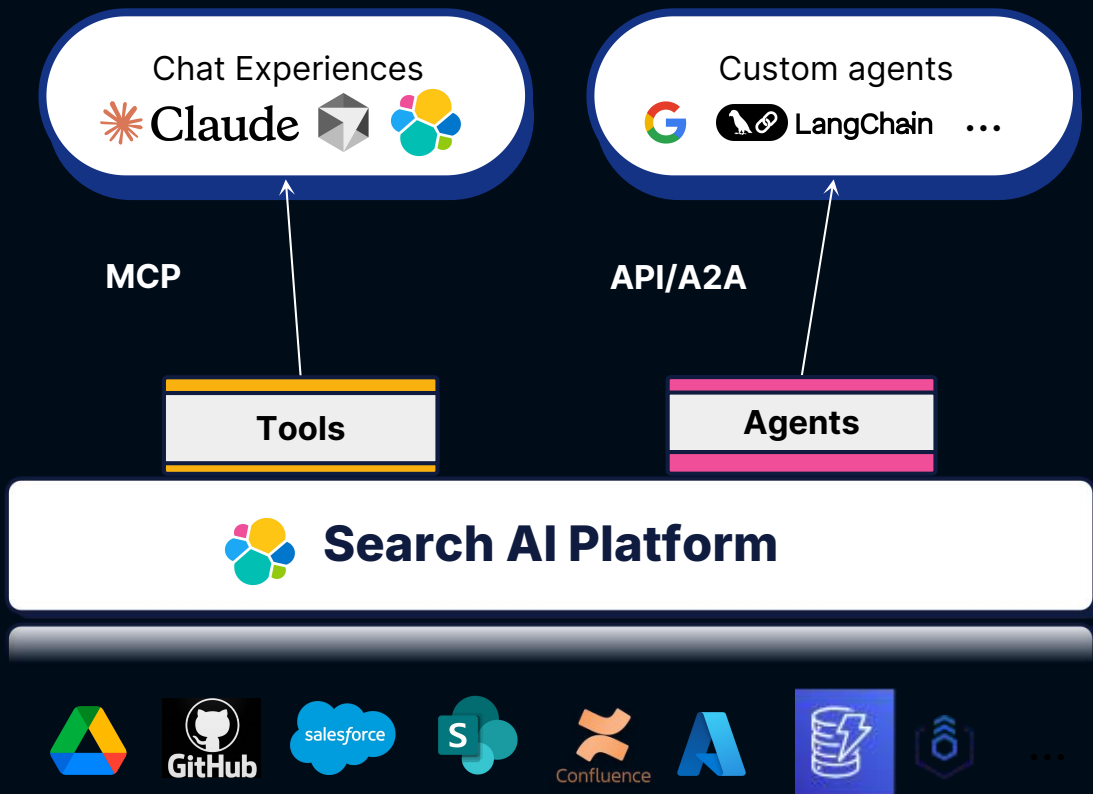


...

Elastic 提供接口，使这些体验轻松实现

DataFun.

AGENTIC AI
超级智能体系架构峰会



搜索比以往更相关

搜索既是
体验



搜索
(结果)

Lexical Search



聊天
(答案)

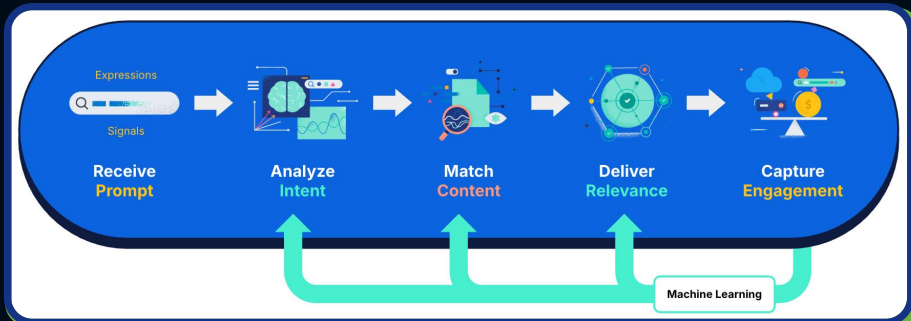
Vector/Semantic/Hybrid Search



Agentic
(自动化)



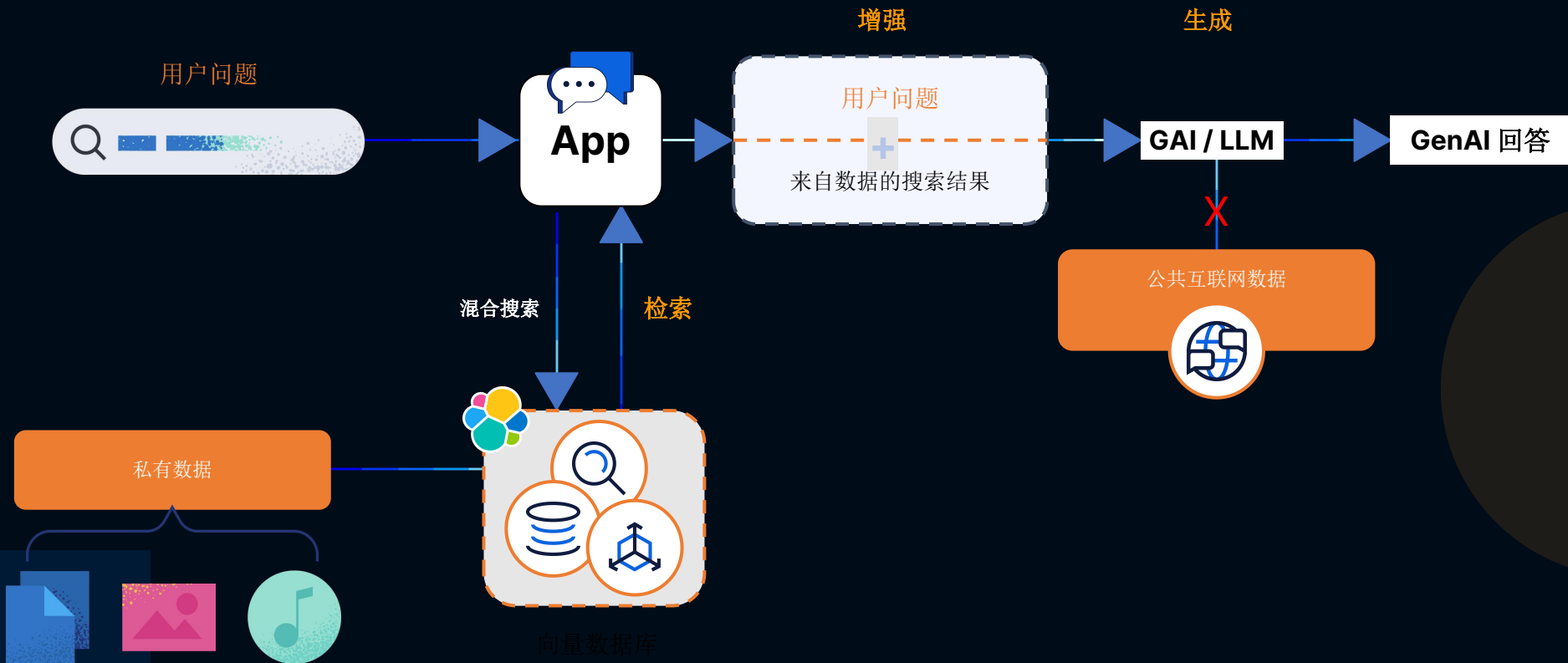
搜索作为
基础



目标

- 理解语义意图并匹配内容
- 提升相关性
- 向 LLM 返回最少量的上下文数据

检索增强生成（RAG）- 消除幻觉



02

Elastic MCP

Elasticsearch + LLM 的力量

想象一下，使用自然语言用英语、葡萄牙语或中文提问关于你的数据的复杂问题，并进行推理

1 Natural Language (MCP)

At what time of day did I take the most steps exploring Las Vegas yesterday?

2 Elasticsearch Query

```
GET apple-health-steps/_search { "size": 0, "query": { "bool": {  
  "must": [ { "match": { "location": "Las Vegas, NV" } }, { "range": {  
    "day": { "gte": "2025-05-13", "lte": "2025-05-13" } } } ] } }, "aggs":  
  { "steps_by_hour": { "terms": { "field": "hour", "size": 24, "order": {  
    "total_steps": "desc" } }, "aggs": { "total_steps": { "sum": { "field":  
      "value" } } } } } }
```

3 SQL (Equivalent)

```
SELECT hour, SUM(value) as total_steps FROM apple_health_steps WHERE  
location = 'Las Vegas, NV' AND day = '2025-05-13' GROUP BY hour ORDER  
BY total_steps DESC LIMIT 1;
```

Elasticsearch MCP Server 当前状态



AGENTIC AI
超级智能体系架构峰会

list_indices:

列出连接集群中的所有索引

get_mappings:

检索指定索引的字段映射

search:

执行完整的 Elasticsearch query DSL 搜索

esql:

在索引上运行 ES|QL（类 SQL）查询。

get_shards:

检查一个或多个索引的 shard 分布和状态

[elastic/mcp-server-elasticsearch](#)

Tools library

Manage MCP ▾

Manage agents

+ New tool

Tools are modular, reusable Elasticsearch operations. Agents use them to search, retrieve, and analyze your data. Use our built-in tools for common operations, and create your own for custom use cases. [Learn more](#)

platform.

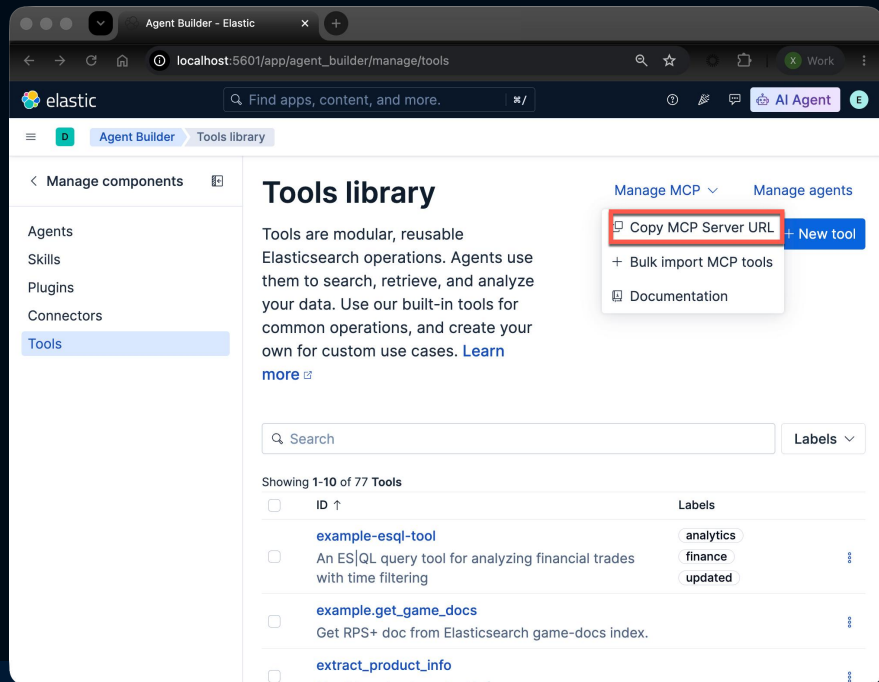
Labels ▾

Showing 1-10 of 77 Tools

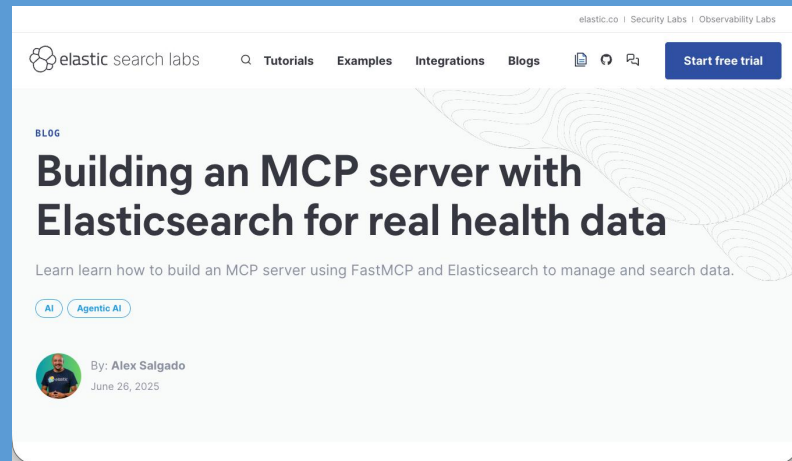
ID ↑	Labels
platform.core.cases	
Retrieves cases from Elastic Security, Observability, or Stack Management. Supports three operation modes:	cases
platform.core.create_visualization	
Create or update a visualization configuration based on a natural language description.	
platform.core.execute_connector_sub_action	
Execute a sub-action on a connector instance. IMPORTANT: Before calling this tool, you MUST read the connector attachment to find the exact sub-action names, required parameters, and the connectorId. Do not guess sub-action names or parameters. The connector attachment lists all available sub-actions with their parameter schemas.	connector sub-action
platform.core.execute_esql	
Execute an ES QL query and return the results in a tabular format.	
platform.core.generate_esql	
Generate an ES QL query from a natural language query.	
platform.core.get_document_by_id	
Retrieve the full content (source) of an Elasticsearch document based on its ID and index name.	
platform.core.get_index_mapping	
Retrieve mappings for indices, aliases or datastreams.	
platform.core.get_workflow_execution_status	
Retrieve the status of a workflow execution.	
platform.core.index_explorer	
List relevant indices, aliases and datastreams based on a natural language query.	

MCP Server + Elastic 的两种方法

定制

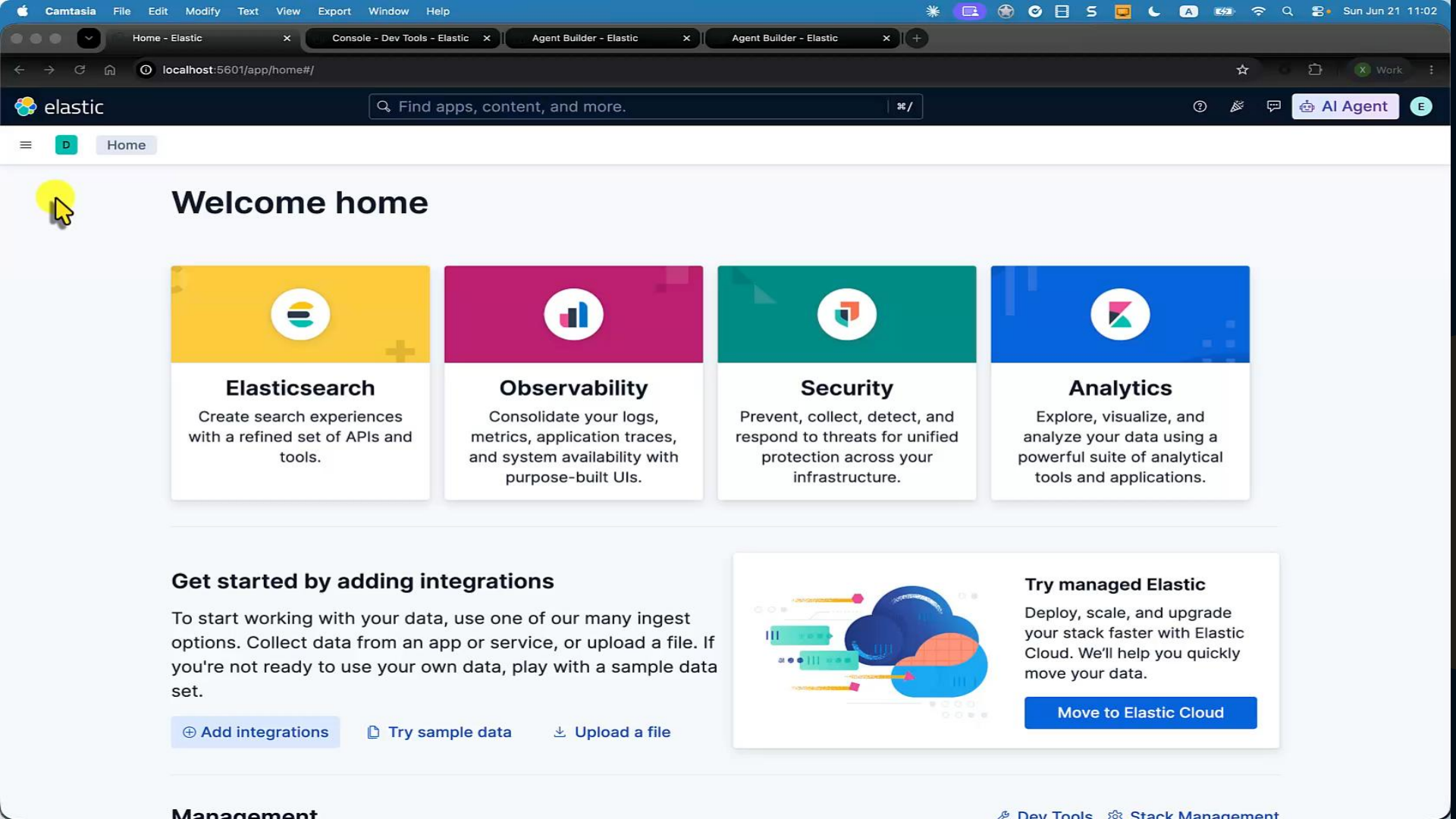


- 领域特定
- 需要编码
- 完全可控



<https://www.elastic.co/search-labs/blog/how-to-build-mcp-server>

K在 Kibana 中集成有一个 MCP Server



Welcome home



Elasticsearch

Create search experiences with a refined set of APIs and tools.



Observability

Consolidate your logs, metrics, application traces, and system availability with purpose-built UIs.



Security

Prevent, collect, detect, and respond to threats for unified protection across your infrastructure.



Analytics

Explore, visualize, and analyze your data using a powerful suite of analytical tools and applications.

Get started by adding integrations

To start working with your data, use one of our many ingest options. Collect data from an app or service, or upload a file. If you're not ready to use your own data, play with a sample data set.

-  Add integrations
-  Try sample data
-  Upload a file



Try managed Elastic

Deploy, scale, and upgrade your stack faster with Elastic Cloud. We'll help you quickly move your data.

Move to Elastic Cloud

03

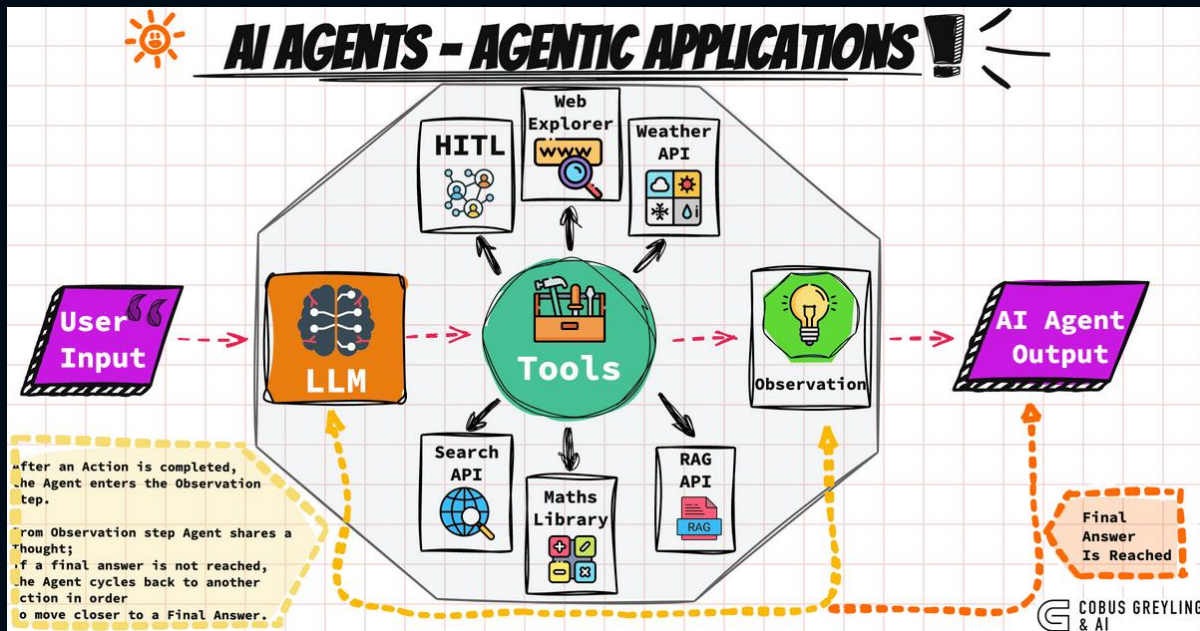
AI agents

曾经幻想过和你的数据对话吗？



什么是 Agent?

AI agent 是一种软件程序，它可以通过做出决策、从其环境中学习，并使用工具来实现目标，从而自主执行复杂任务



Credit: What's Your Definition Of An AI Agent? | Cobus Greyling on Medium

上下文窗口

LLM 一次能够处理的最大 token 数量

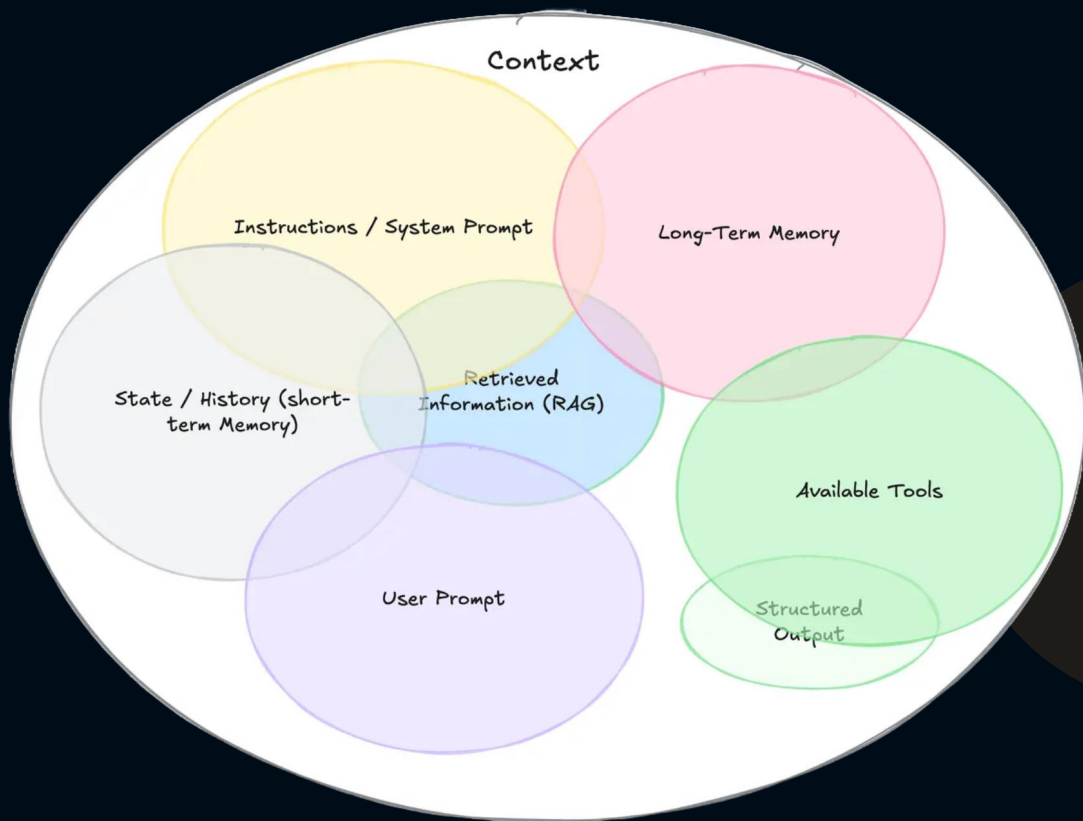
- **信息过少**导致幻觉或错误响应。当 LLM 缺乏足够上下文时，无法确定语义上下文来生成准确响应
- **信息过多导致上下文溢出**。它会压垮 LLM 的注意力范围，降低整个上下文窗口的相关性，使模型难以判断哪些部分最重要
- **分散或冲突**信息会让模型混淆。更大的上下文窗口增加了出现冲突或无关信息的机会，从而干扰 LLM 回答



什么是上下文工程?

上下文工程确保 agents 拥有正确的信息

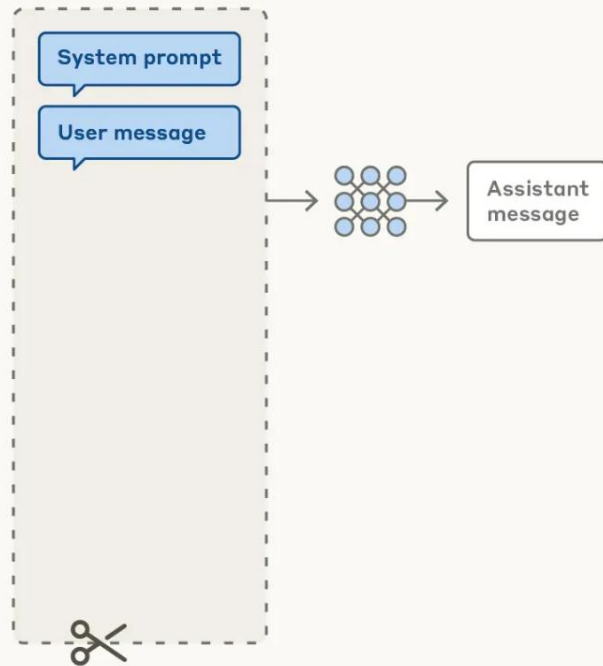
- 垃圾进, 垃圾出
- 搜索没有消失, 它只是换了位置



Prompt engineering vs. context engineering

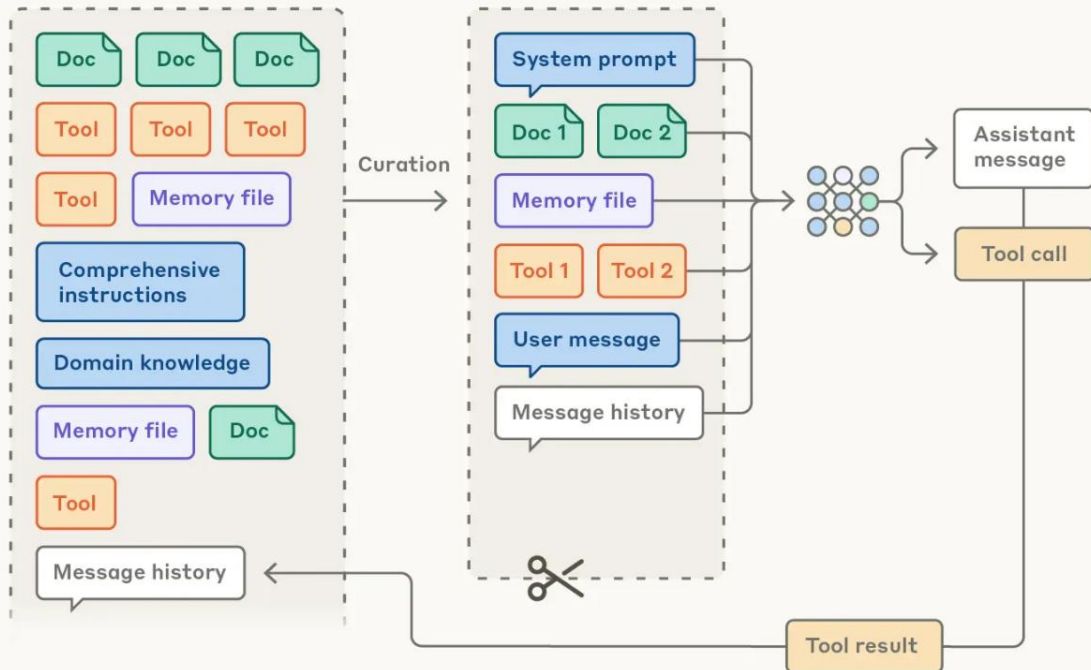
Prompt engineering for single turn queries

Context window



Context engineering for agents

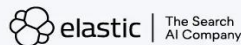
Possible context to give model



混合搜索是各种方法中最优的

Hybrid search is the best of all worlds

First-stage retrieval that *includes* subjective relevance



SQL

"Order by"

Search Engines

Relevance
scoring

Filtering
Aggregations
Ranges

Vector Search

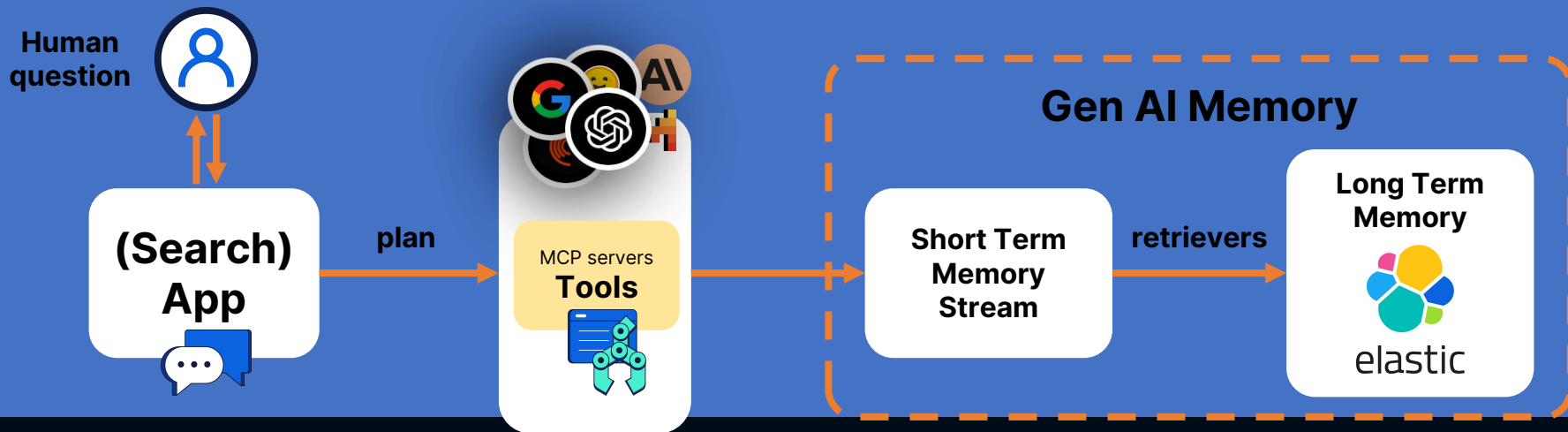
Nearest neighbor

Reranking

Reciprocal Rank Fusion (RRF)
Semantic Re-ranking
Learn-to-Rank (LTR)

```
GET my-hybrid-index/_search {
  "size": 20,
  "retriever": {
    "text_similarity_reranker": {
      "retriever": {
        "rrf": {
          "retrievers": [
            {
              "standard": {
                "query": {
                  "multi-match": {
                    "query": "machine learning applications",
                    "fields": ["title", "content"]
                  }
                },
                "sort": [
                  { "_score": { "order": "desc" } },
                  { "popularity": { "order": "desc" } }
                ]
              },
              "semantic": {
                "field": "semantic_content",
                "query_text": "machine learning applications",
                "model_id": ".elser-2-elasticsearch",
                "k": 50
              }
            },
            {
              "rank_window_size": 50,
              "rank_constant": 20
            }
          ]
        },
        "field": "content",
        "inference_id": "my-reranker",
        "inference_text": "machine learning applications",
        "rank_window_size": 20
      }
    }
  }
}
```

新兴的 agentic 模式

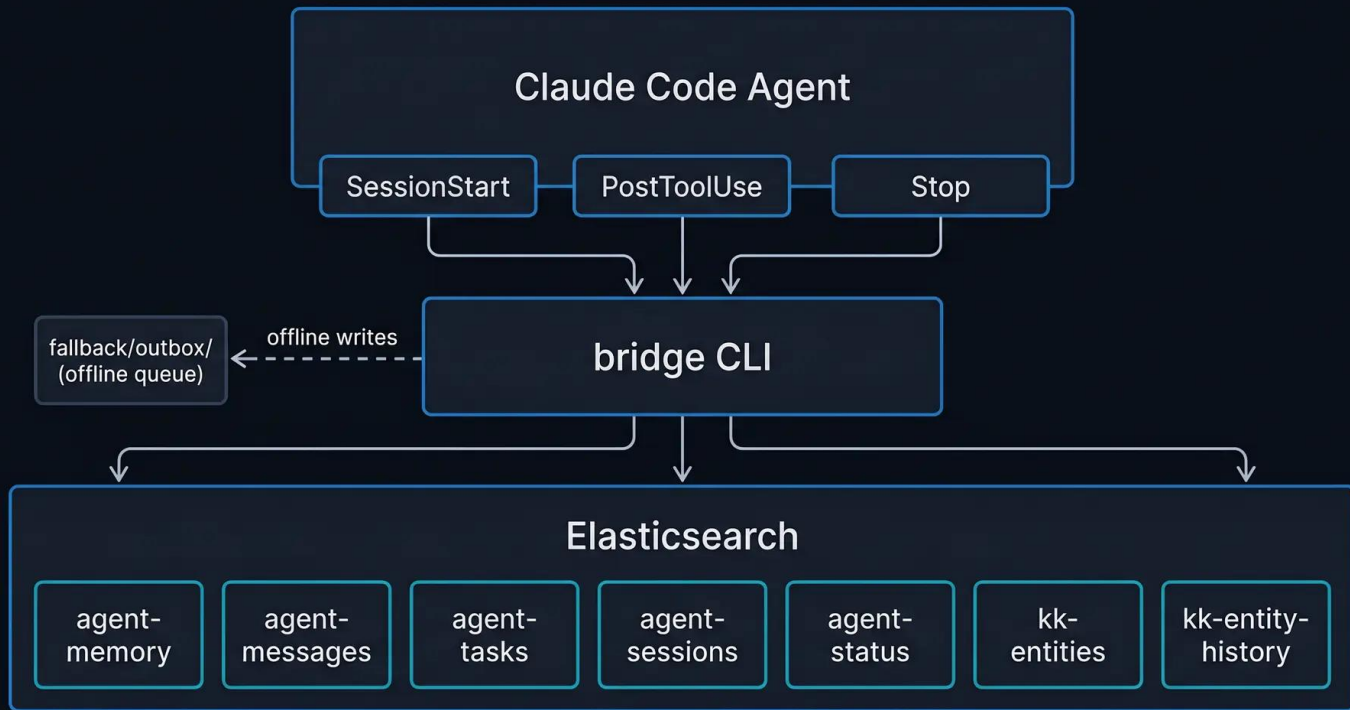


使用 Elasticsearch 管理 agentic 记忆

为什么 Elasticsearch 适合做记忆？

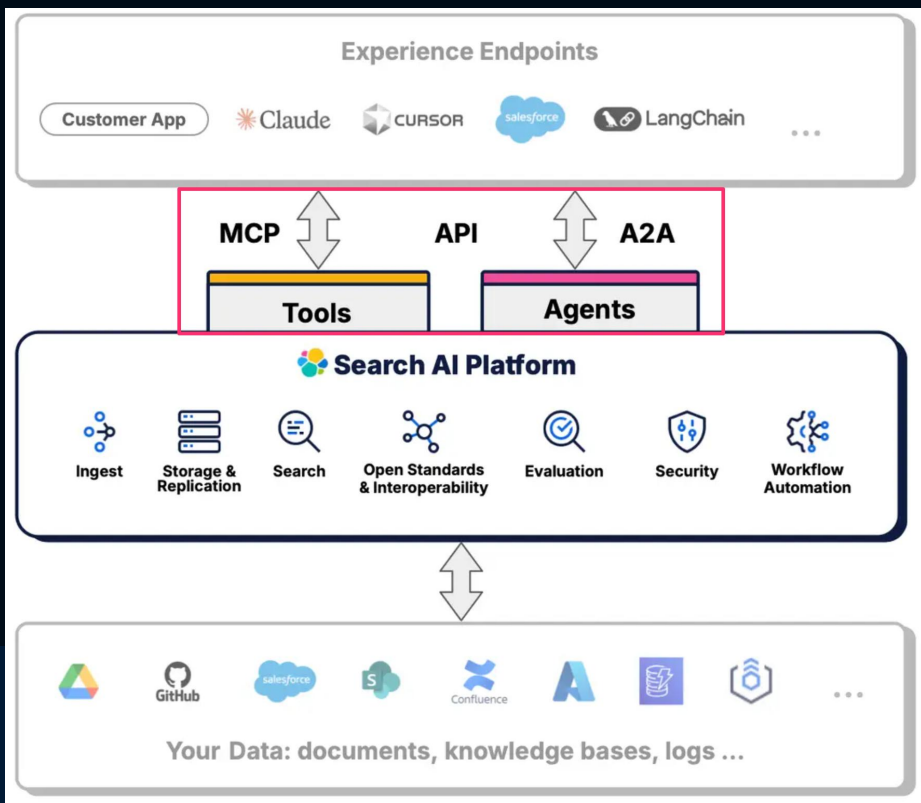
- **开箱即用的混合搜索**。Elasticsearch 结合了 BM25 词法评分与密集的 vector search（指向 inference point 的 semantic_text 字段）
- **通过 ES|QL 实现查询表达能力**。在 Elasticsearch 中，你的 memory 是可以通过完整查询语言访问的文档。ES|QL 支持按类型、日期范围、agent ID、访问范围或任意组合进行过滤，同时支持聚合、时间函数以及 FUSE 多 retrieval 融合。ES|QL 是一门完整的查询语言，具备聚合与时间函数能力，这比大多数 vector store 提供的轻量级 metadata 过滤更强
- **在语义评分之前进行 metadata 过滤**。按 agent 作用域隔离 memory、时间窗口过滤、基于类型过滤，这些都是 Elasticsearch 的标准查询能力。你不需要构建自定义逻辑，只需要组合已有原语。
- **通过 DECAY 实现时间衰减**。系统会对最近的 memory 赋予更高权重，使其排名高于过时内容
- **你已经具备的运维能力**。监控、告警、索引生命周期管理、备份。这些在你的 Elasticsearch 部署中都已经存在

Agent-memory 架构



在 Elasticsearch 上构建一个持久 agent 记忆层 - MCP Server

什么是 Agent Builder?



Elasticsearch 中的一个新 AI 层，提供构建 AI agentic 工作流的框架，使用混合搜索为 agents 提供其推理和行动所需的上下文。

Elastic 将知识暴露给 Agentic 生态系统

DataFun.

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AI 原生体验

Chat Assistants, Voice Assistants, AI Reasoning

Custom Apps



Claude



salesforce



LangChain

...

由工具与 agents 驱动

MCP

A2A

Tools

Agents

由平台支持

Search AI Platform



Ingest



Process



Storage & Replication



Search



AI & ML Analysis



Visualization



Workflow Automation

Search AI Lake

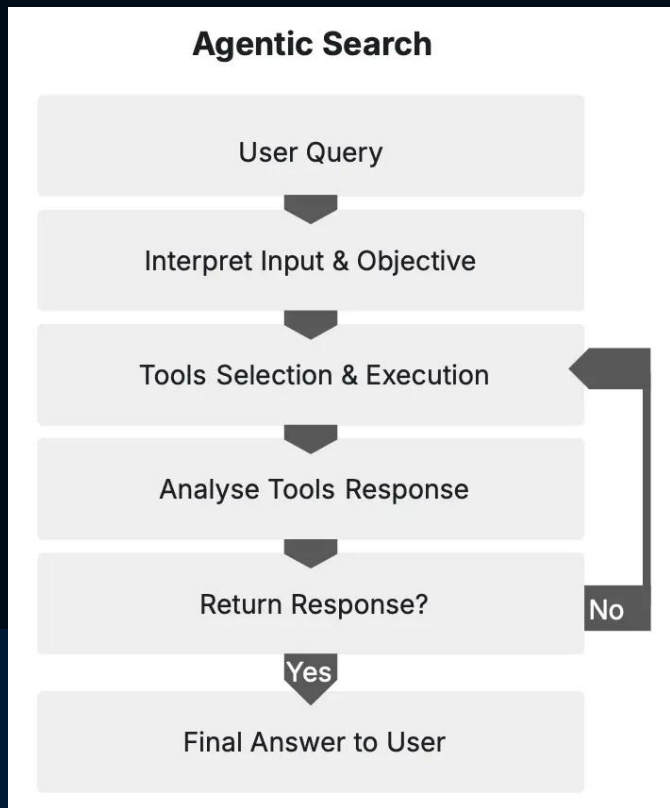
构建于
企业数据



...

Enterprise Data Sources

Elastic Agent Builder 功能



使用原生对话式 **agent** 与你的数据聊天，并查看完整的步骤和结果追踪



利用内置工具查找相关索引，理解数据结构，并将自然语言翻译为结构化的语义、词汇或混合查询



使用 **ES|QL/MCP/Workflow/Index search** 构建自定义工具，以供你自己的 **agents** 使用



内置聊天可视化，支持可编辑图表类型，并可添加到仪表板

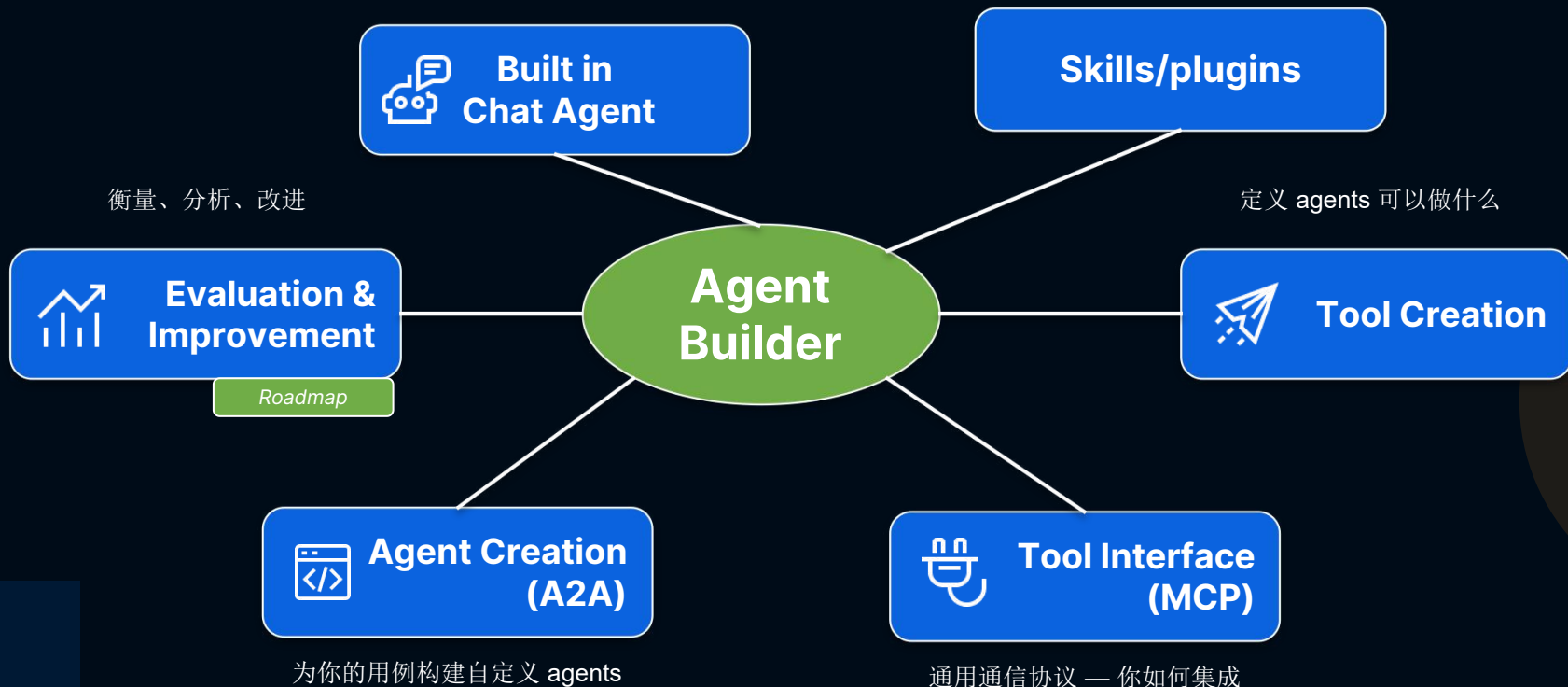


使用 **MCP** 和 **A2A** 与外部 **agents** 和应用集成

Elastic Agent Builder 的六大要素

开箱即用的对话式 agent

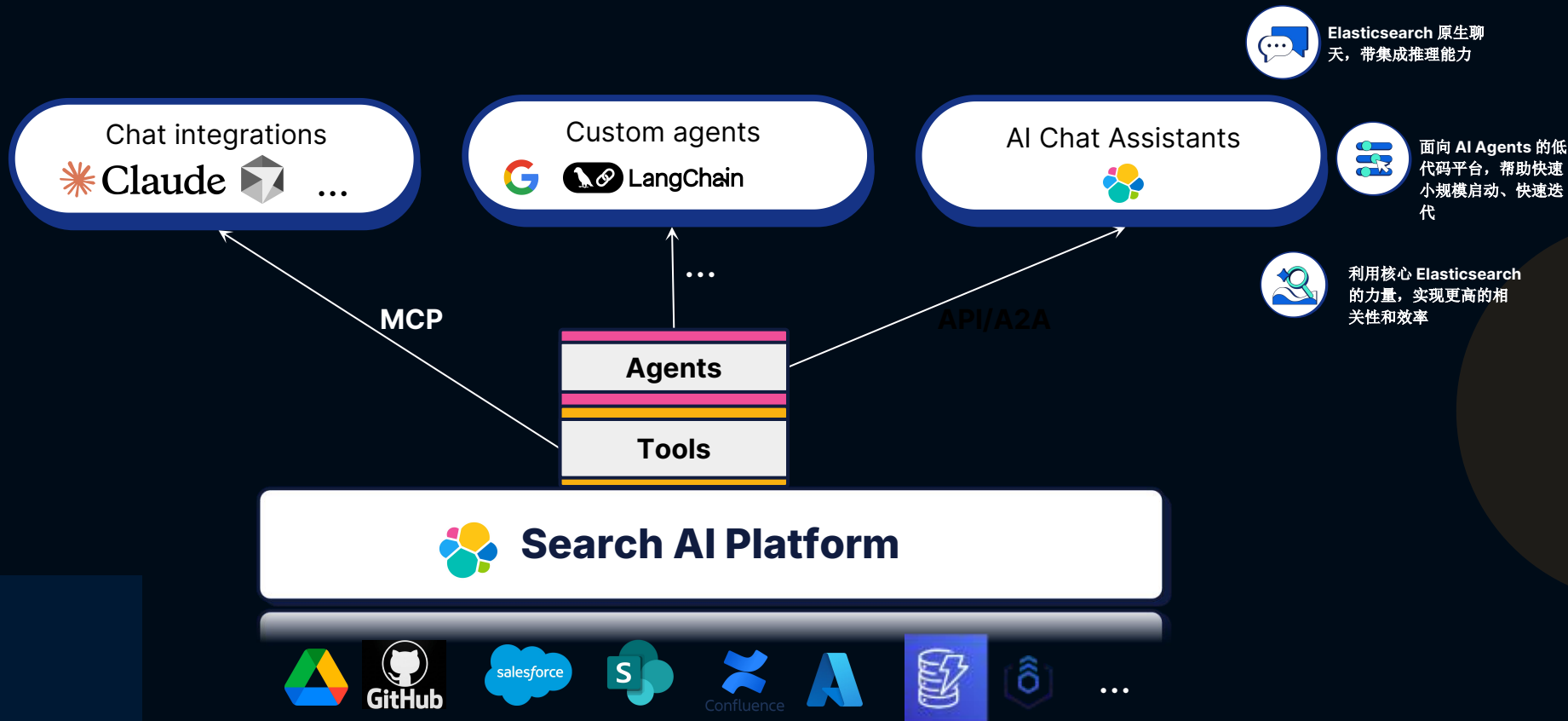
可复用的指令和工具集



Elastic Agent Builder 让这一切变得简单



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内置工具

当前有多个可用工具，前缀为
platform.core:

执行 ES|QL 查询

从自然语言生成 ES|QL 查询

根据 ID 获取完整文档内容

检索一个或多个指定索引的映射

根据自然语言查询列出相关索引、
别名和数据流

列出集群中的索引、别名和数据
流

关键搜索工具

Showing 1-10 of 49 Tools

ID ↑	Labels
platform.core.cases	
Retrieves cases from Elastic Security, Observability, or Stack Management. Supports three operation modes:	cases
platform.core.execute_esql	
Execute an ES QL query and return the results in a tabular format.	
platform.core.generate_esql	
Generate an ES QL query from a natural language query.	
platform.core.get_document_by_id	
Retrieve the full content (source) of an Elasticsearch document based on its ID and index name.	
platform.core.get_index_mapping	
Retrieve mappings for the specified index or indices.	
platform.core.get_workflow_execution_status	
Retrieve the status of a workflow execution.	
platform.core.index_explorer	
List relevant indices, aliases and datastreams based on a natural language query.	
platform.core.integration_knowledge	
Search and retrieve knowledge from Fleet-installed integrations. This includes information on how to configure and use integrations for data ingestion into Elasticsearch.	integration knowledge-base fleet
platform.core.list_indices	
List the indices, aliases and datastreams from the Elasticsearch cluster.	
platform.core.product_documentation	
Search and retrieve documentation about Elastic products (Kibana, Elasticsearch, Elastic Security, Elastic Observability).	
platform.core.search	
A powerful tool for searching and analyzing data within your Elasticsearch cluster.	

创建 tools 的四种方式

The screenshot shows the Elastic Agent Builder interface for creating a new tool. The top navigation bar includes the Elastic logo, a search bar, and an AI Assistant button. The main header shows the breadcrumb 'Agent Builder > Tools > Create a new tool'. The title 'Create a new tool' is prominently displayed with 'Save & test' and 'Save' buttons. A descriptive paragraph explains the need for a unique ID and clear description. The 'Type' section features a dropdown menu with 'ES|QL' selected, and a list of other options: 'Index search', 'Workflow', and 'MCP'.

elastic Find apps, content, and more. ? AI Assistant E

Agent Builder Tools Create a new tool

Create a new tool

Save & test Save

Give your new tool a unique ID and a clear description, so both humans and LLMs can understand its purpose. Add labels for organization, and write the index pattern or ES|QL query that powers its functionality. [Learn more](#)

Type

Set the tool's type and the parameters that control how it operates.

[Documentation - Configuring a tool](#)

Type

ES|QL

- ✓ ES|QL
- Index search
- Workflow
- MCP

Elastic Workflows - YAML 结构

```
name: National Parks Demo
description: Creates an Elasticsearch index, loads sample national
park data, searches for parks, and displays the results.
enabled: true
consts:
  indexName: national-parks

triggers:
  - type: manual

steps:
  - name: get_index
    type: elasticsearch.indices.exists
    with:
      index: '{{ consts.indexName }}'

  - name: check_if_index_exists
    type: if
    condition: 'steps.get_index.output: true'
    steps:
      - name: index_already_exists
        type: console
        with:
          message: 'index: {{ consts.indexName }} already exists.
Will proceed to delete it and re-create'
      - name: delete_index
        type: elasticsearch.indices.delete
        with:
          index: '{{ consts.indexName }}'
    else:
      - name: no_index_found
        type: console
```

● name / description / tags

用于标识和搜索的元数据

● consts

在整个工作流中复用的固定值。通过 {{ consts.name }} 访问

● inputs

每次执行时变化的参数。由用户在运行时提供。

● triggers

工作流如何启动：手动、定时、告警或 webhook。

● steps

内部动作：在 Elasticsearch 和 Kibana 内执行的操作，例如查询索引、运行 Elasticsearch Query Language (ES|QL)、创建 case 或更新告警。

外部动作：在外部系统上执行的操作，例如发送 Slack 消息或创建 Jira 工单。你可以使用在 Elastic 中配置好的任何 connector，也可以通过 HTTP 步骤灵活调用任意 API 或内部服务。

流程控制：使用条件、循环和并行执行来定义工作流逻辑。

AI：从向大语言模型 (LLM) 提示，到将 agent 作为工作流步骤启用，解锁 agentic 工作流用例。

一个 agent 的组成部分

The screenshot displays the Elastic Agent Builder interface. At the top, the Elastic logo and a search bar are visible. The main navigation bar includes 'Agent Builder', 'Agents', and 'property_search_skill'. On the left, a sidebar lists 'Manage components' with sub-items: Agents, Skills, Plugins, Connectors, and Tools. The main content area is titled 'Property search skills' with a sub-header 'Search for property'. Below this, a red box highlights the 'Settings' tab, which is selected. Other tabs include 'Tools' (6), 'Skills' (1), and 'Plugins' (0). The 'Settings' section is divided into two columns. The left column contains 'System references' (describing the agent's behavior) and 'Agent ID' (unique ID for code reference). The right column contains 'Custom Instructions' (describing the agent's role as an information extraction assistant) and a 'Preview' button. The 'Agent ID' field is set to 'property_search_skill'.

elastic Find apps, content, and more. %/ ? AI Agent E

≡ D Agent Builder Agents property_search_skill

< Manage components

Agents Skills Plugins Connectors Tools

PS Property search skills Owner: elastic Chat Save

Search for property

Settings Tools 6 Skills 1 Plugins 0

⚙️ System references Used behind the scenes to identify and guide the agent's behavior. Not shown to end users.

Agent ID property_search_skill

Custom Instructions Optional Preview

This agent is used to search for properties:

Step 1:
You are an information extraction assistant.
Extract real estate search parameters from the user query.

Camtasia

File

Edit

Modify

View

Export

Window

Help

Agent Builder - Elastic

Console - Dev Tools - Elastic

Agent Builder - Elastic

Agent Builder - Elastic

Send Message to WeCom

LlamaParse Extract Economic

Frontier Market Economic

localhost:5601/app/dev_tools#/console/shell

Find apps, content, and more.

%/

AI Agent

elastic

Dev Tools

Console

Console

Search Profiler

Grok Debugger

Painless Lab

BETA

Shell

History

Config

Export requests

Import requests

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GET _cat/indices/finan*

GET financial_holdings/_search

GET financial_reports/_search

GET financial_asset_details/_search

GET financial_accounts/_search

GET financial_news/_search

POST /_bulk

{ "index" : { "_index" : "people", "_id" : "1" } }

{ "id": 1, "name" : "John Doe", "description" : "A software developer", "sex" : "Male", "age" : 30, "address" : "123 Elm Street, Springfield", "location": {"lat": 37.7749, "lon": -122.4194}, "date_of_birth": "1995-05-15" }

{ "index" : { "_index" : "people", "_id" : "2" } }

{ "id": 2, "name" : "Jane Smith", "description" : "A project manager", "sex" : "Female", "age": 28, "address": "456 Maple Avenue, Anytown", "location": {"lat": 40.7128, "lon": -74.0060}, "date_of_birth": "1997-03-22" }

{ "index" : { "_index" : "people", "_id" : "3" } }

{ "id": 3, "name" : "Alice Johnson", "description" : "A graphic designer", "sex": "Female", "age": 26, "address": "789 Oak Lane, Metropolis", "location": {"lat": 34.0522, "lon": -118.2437}, "date_of_birth": "1999-07-08" }

{ "id": 4, "name" : "Bob Brown", "description" : "A data analyst", "sex": "Male", "age": 35, "address": "101 Pine Street, New York", "location": {"lat": 40.7128, "lon": -74.0060}, "date_of_birth": "1990-11-11" }

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This resolution is viewed as a major win, showcasing the resilience of the ETF's strategy in navigating complex international legal landscapes. Analysts believe the positive outcome will likely reinforce investor trust in the fund's broad market approach, potentially attracting new capital from those who had been wary of the legal risks. The news underscores the benefits of diversification in mitigating single-company controversies."

{ "source": "Bloomberg", "published_date": "2026-03-20T09:41:27", "url": "http://fakenews.com/article/8c65ff41", "entities": [{ "symbol": "VXUS", "name": "Vanguard Total International Stock ETF", "type": "ETF" }], "sentiment": "positive", "last_updated": "2026-03-20T09:41:27", "primary_symbol": "VXUS", "company_symbol": null }

{ "_index": "financial_news", "_id": "72c3df8a-c98d-4ee3-9d6d-1db4b32959de", "score": 1 }

Clear this input

Clear this output

200 - OK

42 ms



THANKS

演讲嘉宾：刘晓国
